

CURRICULUM VITAE

Sabina Isgandarova, PhD

Chief Sustainability Officer

E2E Energy Solutions, Canada

Contact and identifiers

Full name (as in publications)	Sabina Isgandarova
Name in Azerbaijani	Səbinə İsgəndərova
Country of residence	Canada
Institutional email	sabina@ualberta.net
Scientific field	Biology; environmental management

Profile summary

Sabina Isgandarova is chief sustainability officer at E2E Energy Solutions, where she has worked since January 2022. She received a PhD (Dr. rer. nat.) in 2004 from the Technical University of Berlin and the Max Planck Institute for Chemical Energy Conversion; the thesis is on photosynthetic water oxidation in cyanobacteria and higher plants. She holds a 1997 diploma in biology and in the teaching of biology and chemistry, with a specialization in microbiology, from Baku State University. She has taught environmental subjects at the University of Calgary since 2013 and at Bay River College from 2019 to 2024, and she is the founder of AlgaePro Energy and Environment Limited.

Education

Years	Degree	Field / thesis title	Institution, city, country
January 2004	PhD	Dr. rer. nat. Thesis: Studies on the mechanism of photosynthetic water oxidation in cyanobacteria and higher plants	Technical University of Berlin and Max Planck Institute for Chemical Energy Conversion, Mülheim an der Ruhr, Germany
July 1997	Diploma equivalent to an MSc	Biology and teaching of biology and chemistry, specialization in microbiology. Thesis: morpho-physiological analysis of microscopic pathogenic fungi in the rhizosphere of plants in oil-contaminated soils	Baku State University, Baku, Azerbaijan

Academic and professional positions

Years	Position	Department / unit	Institution, city, country
January 2022–present	Chief Sustainability Officer		E2E Energy Solutions
May 2019–October 2024	Instructor	Applied Environmental Technologies	Bay River College
2013–present	Instructor	Environmental Management Certificate program, Faculty of Continuing Education	University of Calgary, Calgary, Canada

Years	Position	Department / unit	Institution, city, country
2013–present	Founder and principal scientist		AlgaePro Energy and Environment Limited
2013–2015	Scientist and process advisor		Canadian Natural Resources Limited
January–December 2014	Consultant, water treatment		Ledcor Group for Applied Research in Oil Sands Environmental Sustainability, Northern Alberta Institute of Technology
2011	Postdoctoral fellow		Umeå University, Umeå, Sweden
2004–2010	Postdoctoral fellow		University of Alberta, Edmonton, Canada

Publications

Books and textbooks

1. **Isgandarova, Sabina.** Photosynthetic water oxidation in higher plants and microalgae. VDM Publishing Group, 2010. ISBN-10 363929906X.

Peer-reviewed journal articles

1. **Isgandarova, S., et al.** Stimulation of actin polymerization by vacuoles via Cdc42p-dependent signaling. *Journal of Biological Chemistry*, 2007, 282(42), 30466–30475.

2. **Isgandarova, S.; Renger, G.; Messinger, J.** Functional differences of photosystem II from *Synechococcus elongatus* and spinach characterized by flash-induced oxygen evolution patterns. *Biochemistry*, 2003, 42, 8929–8938.

Awards, honors and fellowships

Year	Award / honor	Awarding body
December 2023	Building Cultural Relationships Medal	Republic of Azerbaijan
October 2019	Instructor of the Month	Bay River College, Calgary
2018	Faces of the Community	Federation of Calgary Communities, Calgary, Canada
2008	Merck-Frosst Postdoctoral Research Award	Edmonton
2008–2010	Alberta Innovates (AHFMR) Postdoctoral Research Fellowship	
2002–2004	Predocctoral and postdoctoral fellowships	Max Planck Society
2011	Wallenberg Foundation Fellowship	Umeå University

Teaching

Years	Course	Level	Institution
2013–present	Introduction to Environmental Management; Environmental Stewardship and Sustainability; Environmental Risk Assessment and Management		University of Calgary

Years	Course	Level	Institution
2019–2024	More than 20 courses in environmental sciences and technologies		Bay River College